

Project Title: Elite Luxury Rental Management System

CS 340 Group 45 Project - Oregon State University

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URL Link: <http://classwork.engr.oregonstate.edu:6875/>

Feedback by the TAs and peer reviewer:

The following are feedback from the ED Discussion for the step 1 draft:

Alexander Croddy

Your overview, overall, does a good job of describing facts and figures relating to the problem as well as the solution, and a realistic scenario requiring an efficient database to solve that has probably been a painful trial of more than one growing business.

Six entities (clients, vehicles, locations, employees, rentals, and employee-client relations) more than fulfill the requirements for storing separate but related ideas, and each one has an effectively-prepared list of important attributes to store as data as well as their relationships. I may suggest as a precaution, however, that an employee have their list of clients or number of currently-served clients added to their own table, preferably using the client-employee relation table as a reference.

The presentation and formatting of the data are all of consistent quality, using camelCase for attributes and PascalCase for table names. Additionally, relationships are very well defined, having both M:M and 1:M relationships with proper formulation, and a description of how and why they are listed as they are with regards to the entities they connect.

In short, the project's looking good, and I expect to see a very well-put-together end product by the last weeks of the term.

Lkham-Erdene Nyamjav

Hey team, it is a really nice project idea—a luxury vehicle rental system is a lot more interesting than the typical store or library database. Here's my feedback:

Your overview does an excellent job describing the problem. The explanation of overlapping bookings (15–20 per month), missed maintenance notifications, and revenue loss from inefficient car usage gives a very concrete picture of why a database solution is needed. It's clear this isn't just a theoretical exercise — it reads like a real business pain point that the system would actually solve.

This is one of the stronger overviews I've seen in terms of specific numbers. A fleet of 85 vehicles, 1,200 clients per year, 2,800 rental deals, \$14 million in annual revenue, and three Southern California branches — that all paints a very clear picture of the scale and justifies a robust database solution. Nothing really missing here.

You have 6 entities — Clients, Vehicles, Locations, Employees, Rentals, and EmployeeClients — all well above the minimum and each representing a clearly distinct concept. I especially liked the Locations entity since it ties everything together across branches. The EmployeeClients intersection table is a smart addition that handles a relationship a lot of groups probably wouldn't think to include.

The attribute detail is thorough and consistent across all entities. Datatypes, constraints, and purposes are all present which is great. One small thing — the Employees table has "Role" capitalized while every other attribute uses camelCase. Minor but worth fixing for consistency. Also worth considering whether membershipTier in Clients should have a default value since not every client may have a tier assigned right away.

Both M:N relationships are clearly identified and properly resolved — Clients to Vehicles through Rentals, and Employees to Clients through EmployeeClients. The 1:M relationships are all correctly formulated. One thing I noticed in the ERD is that the dark background makes it a bit hard to read at a glance — a lighter theme would make it easier for graders to follow. Also the Rentals box in the ERD shows "Rentals" as an attribute name inside the box which looks like it might be a label artifact rather than an actual attribute.

Naming is mostly consistent with camelCase for attributes and PascalCase for tables. Tables are all plural except "Locations" which appears to have "Locations" listed again as an attribute inside the ERD box — that looks like it might be a duplicate label worth cleaning up. The outline and overview are consistent with each other in terms of entity names which is good.

Overall this is a really well put together draft. The overview is strong, the schema is thoughtful, and the two M:N relationships are both well justified. Main things to polish before the final version would be the ERD readability and a couple of minor naming inconsistencies.

Natasha Ritter

I like the story that comes with your database, it gives straightforward information about the number of cars available, the estimated stats regarding rentals and deals annually. Something I did notice, though your database is very detailed already in each section, is that the presentation includes companies having a problem with missed maintenance. I may have missed it but I didn't see anything related to that in the outline, I do see the availability status for the vehicle which is great, but nothing with regard to maintenance. Something that could substitute having a whole other section is just including due for maintenance in your options for availability just to touch on what's in the plan portion. It looks great, and I wish my insurance covered luxury car rentals :).

Facts:

Facts in this presentation show the company inventory, what I believe is a good estimate of clients and rentals annually, and enough components to warrant a schema database system for sure. Especially to get away from paper and excel sheets, I've seen so many hiccups in businesses with these forms of operation. I can see why the overlapping takes place.

Entities/Attributes:

You list very clearly the six entities for your presentation: Clients, Vehicles, Locations, Employees, Rentals, EmployeeClients. All six qualify as relevant information with regard to preventing mishaps that have occurred without a database. Each one has a stated purpose; an outline of variables that will help identify and cross reference within the database without confusing elements. If locations are interchangeable (as I have seen and done by picking up in one spot and dropping off at another), I would suggest adding something to the locationID so customers don't come to the wrong location or get scheduled for a car that isn't there.

Relationships & ERD:

I don't see any standout issues with the plan, relationships all look appropriate. You have clients to rentals as 1:M, clients to vehicles are M:M, employees to clients are M:M, and locations to vehicles are 1:M. The structure matches what the list presents very well.

Consistency:

The Employee section has one element that needs to be adjusted and that's the Role. It shows up as role mostly, and I believe we have to keep the capitalization consistent for most of our coding if not all. Otherwise everything else looks really good!

Statement:

My opinions in this review are my own, and my outline was formatted based off of the assignment Project Step 1 Review on Canvas.

Quan Doan

Hello Group 45!

Does the overview describe what problem is to be solved by a website with DB back end? If yes, summarize. If not, what changes would better support describing the problem to be solved?

Yes, the overview describes what problem is to be solved by a website and DB backend. It talks about a rental management system for luxury cars. It will be used for all processes related to recording the availability of cars, customer preferences, booking deals, and car maintenance, and missed notifications about need for maintenance.

Does the overview list specific facts? If yes, summarize what the facts illustrate about the proposed DB solution. If not, what facts would better support illustrating the scope and scale of the proposed DB solution?

Yes, the overview lists specific facts. The firm has a fleet of 85 luxury cars and will serve about 1,200 rich clientele each year. The company makes more than 2,800 rental deals every year from 3 branches in SoCal. They will use a database-oriented admin web portal to help companies handle their vehicles, clients, employees, locations, and rental details.

Are at least four entities described, and does each one represent a single idea to be stored as a list? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

Yes, there are a total of 6 entities. The entities are Clients, Vehicles, Locations, Employees, Rentals, and EmployeeClients.

The Clients record details of individuals who rent vehicles. The Vehicles are for storing information about each luxury vehicle in the rental fleet. The Locations represent the company's rental branches/offices. The Employees record details of staff members who manage rentals and interact with clients. The Rentals for implementing M:N relationship between Clients & Vehicles, while recording rental transaction details. The EmployeeClients is a table that implements the M:N relationship between Clients and Employees.

Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints, and describe relationships between entities? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

Yes, the outline of entity details describe the purpose of each, list attribute datatypes and constraints, and describe relationships between entities. For example, the ids in all the entities have int, auto_increment, unique, not NULL, PK. For varchar, the properties are firstName, lastName, email, phone, make, model, licensePlate, status, locationName, address, city, phone, and role. Date is also used for all date attributes.

Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

The 1:M relationships are correctly formulated. For example, there are Clients to Rentals, Vehicles to Rentals, Locations to Rentals, Employees to Locations, and Vehicle to Locations.

There is at least 1 M:N relationship. For example, there are Clients to Vehicles.

The ERD does present a logical view of the database.

Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

a. Yes, there is consistency.

b. Entities are plural and attributes are singular.

c. I noticed Role is capitalized in Employees; however, as long as we are consistent, I think it is fine.

Actions based on the feedback:

- Updated the ERD to include proper crow's foot notation to improve clarity and align with standards.
- Corrected naming inconsistency by changing "Role" attribute in the Employees table to "role" to maintain camelCase formatting.
- Reviewed entity relationships to ensure that the relationships of Clients-Vehicles and Employees-Clients were clearly defined and implemented.
- Reviewed and made sure vehicle maintenance is represented through the status attribute.

Upgrades to the Draft 1 version:

- ERD was edited to add proper crow's foot notation for clearer representation of relationships.
- "Role" was changed to "role" to reflect proper case style.

The following are feedback from the ED Discussion for the step 2 draft:

Lon Danna

Hey Group 45. I've taken a look at your documentation and SQL script. I really like the concept of a high end vehicle rental company.

Your team did a great job with consistency. The entities and attributes in the database outline match your ER diagram. Your entities are plural and your attributes are singular. I notice you made the change from Role to role per peer feedback. Your schema is very easy to read. The choice to use crows feet makes the cardinality much clearer.

Your M:N relationship is handled perfectly with the EmployeeClients intersection table. Your explanation of the 3NF process was very helpful. The choice of data types are solid. Using decimal(10, 2) for dailyRate and totalCost is a good way to avoid rounding errors.

The DDL.sql file appears to have the correct syntax. I like how your team used SET FOREIGN_KEY_CHECKS=0 at the top of the script. This helps ensure the script runs smoothly. PKs and FKs are defined clearly. All of the sample data in your document are present in the INSERT statements. It's enough data to test the relationships. The SQL is very well organized. It feels like your team didn't just do forward engineering in MySQL instead of a messy SQL dump. Good job!

Overall, this is a very well done step 2. If I could make one suggestion, it would be to add a Ducati Panigale V4 to vehicle inventory. Best of luck to your team.

This review was my own work.

- Lon

Jonathan Marchand

Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?

Yes. The schema correctly shows all 6 tables; Locations, Vehicles, Employees, Clients, Rentals, EmployeeClients. Correctly displays the proper primary and foreign keys for each table. All attributes are correctly defined in the outline and cardinality between the tables is correctly expressed.

One thing I would fix is the cardinality between Locations and Vehicles. This is a one to many relationship and the crow's foot should be on the Vehicles table. It kind of looks like it is but it's unclear. You may want to adjust the connection for clarity.

Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

The naming convention is consistent throughout utilizing camelCase for all attributes. Tables are correctly named with the first letter capitalized and made plural. Attributes are correctly made singular. I don't see any issues here.

Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?

Yes, the schema is easy to read. There is just one point I made above that the cardinality between Locations and Vehicles is a little unclear. I can make out the crow's foot on the Vehicles table but others may not. Could create confusion if later referenced.

Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?

Yes, each table is properly formed. I have no suggestions for improvement here.

Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?

Each table utilizes one Primary Key so partial dependencies cannot exist. I do not see any transitive dependencies at this time as the sample data is relative only to its particular table's primary key.

Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)

Ran the sql file through vscode and didn't encounter any errors. All tables created and all sample data correctly populated to each table.

In the SQL, are the data types appropriate considering the description of the attribute in the database outline?

The data types are appropriate for each attribute described in the database outline:

- Primary keys and foreign keys properly use INT
- Names, emails, and phone numbers all correctly use VARCHAR
- Date attributes such as joinDate correctly use DATE

In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?

The primary and foreign keys are correctly defined. Each table has 1 primary key and each foreign key references the correct table. I don't see any rogue keys in any table.

Each CASCADE operation is appropriate. There are 7 total:

- Vehicles to Locations
- Employees to Locations
- Rentals to Clients, Vehicles, Locations (3 total)
- EmployeeClients to Employees and Clients (2 total)

I can't see where else you would need another CASCADE operation. What you have currently is sufficient.

In the SQL, are relationship tables present when compared to the ERD/Schema?

There is only one relationship table, EmployeeClients. It is present in the SQL and displayed on both the ERD and the Schema

In the SQL, is all example data shown in the PDF INSERTED?

All the sample data from the pdf was inserted in the schema. Nothing is missing from what I can see.

Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?

The commenting is sufficient. You could explain the relationships but that would be overkill as they are already defined in your outline. The SQL looks great.

This work is my own. No AI was used in my review.

Jason Hudson

Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?

Yes. The schema model presents a physical model that follows the outline and diagram. However, the 'Rentals' is treated as both a transaction and intersection table.

Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

The naming conventions are consistent. The attributes use the preferred format of plural names and camelCase format.

Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?

Yes, the schema is easy to read—the relationship lines could be spaced further apart and the diagram could be bigger. But, overall it is easy to read.

Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?

The intersection tables are properly formed. 'EmployeeClients' lists two foreign keys and a constraint and 'rentals' connects 'Clients' and 'Vehicles' by using foreign keys and attributes for transactions.

Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?

Yes, the data is separated into distinct tables—client, vehicle, location, employee, rental and employee-client are all separate tables.

Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)

Yes, the foreign key definitions, insertions, and transaction controls are all properly formatted.

In the SQL, are the data types appropriate considering the description of the attribute in the database outline?

Yes, the IDs use the appropriate attributes—money uses the decimal format, date uses the date format, and the text field uses the varchar.

In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?

Yes, the primary keys and the foreign keys match the schema. 'ON UPDATE CASCADE' is appropriate. 'ON DELETE RESTRICT' is fine. 'EmployeeClients' would benefit from using 'ON DELETE CASCADE' for both of the foreign keys.

In the SQL, are relationship tables present when compared to the ERD/Schema?

Yes, the 'EmployeeClients' is present and the 'Rentals' is present to demonstrate the 'Client-Vehicles' relationship table.

In the SQL, is all example data shown in the PDF INSERTED?

Yes, the SQL inserts three locations, four clients, five vehicles, four employees, five rentals, and seven employee-client rows

Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?

Yes, it is well structured, hand-authored, and organized by sections.

This review was my own work.

Garath May

Hi Group 45!

Overall your draft and SQL look solid! A luxury vehicle rental company is a smart choice for building a database around. I'll go ahead and fill out the questionnaire below:

Does the schema present a physical model that follows the database outline and the ER logical diagram exactly?

The schema presents a model that follows the six entities and each of their corresponding relationships. There are all the correct associations and such, unless I'm missing something.

Is there consistency in a) naming between overview, outline, ER and schema entity/attributes b) entities plural, attributes singular c) use of capitalization for naming?

Yes, save for employeeclients in the outline versus EmployeeClients in the SQL. All in all an easy fix and something that you all probably missed. No biggie, but since it is an intersection table, it definitely needs to be fixed.

Is the schema easy to read (e.g. diagram is clear and readable with relationship lines not crossed)?

Yes everything is easy to read and understand, there's a lot of info on those tables! There has to be with so many entity relationships, and you guys seemed to handle them well.

Are intersection tables properly formed (e.g. two FKs and facilitate a M:N relationship)?

Yes, EmployeeClients has two foreign keys (employeeID and clientID) and breaks down the pre-existing M:N relationship effectively.

Does the sample data suggest any non-normalized issues, e.g. partial dependencies or transitive dependencies?

All the data is relative to the tables, and the naming is correct. There seems to be no normalization issues apparent at this time.

Is the SQL file syntactically correct? This can be easily verified by using PhPMyAdmin and your CS 340 database (do not forget to take backup of your own database before you do this!)

Yes the SQL file is syntactically correct, and does not return any errors.

In the SQL, are the data types appropriate considering the description of the attribute in the database outline?

AUTO-INCREMENT is used appropriately, it's are it's and decimals are decimals. Everything seems to be good!

In the SQL, are the primary and foreign keys correctly defined when compared to the Schema? Are appropriate CASCADE operations declared?

All the primary keys are labeled and represented correctly in the SQL, same for the foreign keys. Specifically in the intersection table, which is where it can get tricky.

In the SQL, are relationship tables present when compared to the ERD/Schema?

Yes, and to harp on the intersection table again: it does the job of what would be a M:N relationship effectively.

In the SQL, is all example data shown in the PDF INSERTED?

Yes, all the data from the tables is inserted correctly. A lot of data.

Is the SQL well-structured and commented (e.g. hand authored) or not (e.g. exported from MySQL)?

From what I can tell, yes. Everything seems to be well structured and commented, especially the drop tables at the beginning along with the foreign key checks.

As a reviewer, clearly describe to what extent your feedback to the team was original (e.g. "all my work") or non-original (e.g. used AI tools per Code Citation Tips).

Everything was written by me (except the questions), Garath May

Actions based on the feedback:

- Clarify the crow's foot between Locations and Vehicles in the ERD/schema.
- Make standardization of capitalization for EmployeeClients.
- Update EmployeeClients foreign key to use ON DELETE CASCADE, instead of ON DELETE RESTRICT.

Upgrades to the Draft 2 version:

- Crow's foot notation between Locations and Vehicles within the Schema was corrected and clarified.
- Capitalization of EmployeeClients was standardized within the outline.
- Reformatted to a uniform style for better readability.

Summary:

Elite Luxury Vehicle Rental is a company that rents out elite exotic and premium vehicles; the firm has a fleet of 85 luxury cars including Ferrari, Lamborghini, Rolls Royce, and Porsche models, and serves about 1,200 rich clientele each year. The company makes more than 2,800 rental deals every year from three branches located in Southern California and brings in revenue worth about \$14 million per annum. At present, all processes related to recording the availability of cars, preferences of customers, booking deals, and car maintenance have been managed through fragmented excel sheets and paper logs, which often lead to overlapping bookings (about 15–20 deals per month), missed notifications about the need for maintenance, and revenue loss due to inefficient car usage. A database-oriented admin web portal will enable company officials to manage vehicles, clients, employees, locations, and rental details.

Database Outline:

- **Clients:** Records details of individuals who rent vehicles. This entity tracks client information, preferences, and rental history to enable personalized service and loyalty management.
 - clientID: int, auto_increment, unique, not NULL, PK
 - firstName: varchar(50), not NULL
 - lastName: varchar(50), not NULL
 - email: varchar(100), not NULL, unique
 - phone: varchar(20), not NULL
 - membershipTier: varchar(20) (e.g., "Standard", "Premium", "VIP")
 - joinDate: date, not NULL

Relationships:

- Clients to Rentals: 1:M (one client can have many rentals over time). Implemented with clientID as FK in Rentals.
 - Crows foot notation: Clients || - Rentals |<
- Clients to EmployeeClients 1:M (one client can interact with many employees). Implemented with clientID as FK in EmployeeClients.
 - Crows foot notation : Clients || - EmployeeClients |<
- **Vehicles:** Stores information about each luxury vehicle in the rental fleet. This entity maintains the luxury fleet inventory with current availability and pricing details.
 - vehicleID: int, auto_increment, unique, not NULL, PK
 - locationID: int, not NULL, FK references Locations(locationID)
 - make: varchar(50), not NULL (e.g., Ferrari, Lamborghini)
 - model: varchar(50), not NULL (e.g., 488 Spider, Urus)
 - year: int, not NULL
 - licensePlate: varchar(20), unique, not NULL
 - dailyRate: decimal(10,2), not NULL
 - status: varchar(20), not NULL (e.g., "Available", "Rented", "Maintenance", "Reserved")

- mileage: int

Relationships:

- Vehicles to Rentals: 1:M (one vehicle can be rented many times, but only one active rental at a time). Implemented with vehicleID as FK in Rentals.
 - Crows foot notation: Vehicles || - Rentals |<
- Vehicles to Locations: 1:M (vehicles are assigned to a home location).
 - Crows foot notation: Locations || - Vehicles |<
- **Locations:** Represents the company's rental branches/offices. This entity tracks where vehicles are based and where rentals are picked up/returned.
 - locationID: int, auto_increment, unique, not NULL, PK
 - locationName: varchar(100), not NULL (e.g., "Beverly Hills", "Newport Beach", "Irvine")
 - address: varchar(200), not NULL
 - city: varchar(50), not NULL
 - phone: varchar(20)

Relationships:

- Locations to Rentals: 1:M (rentals occur at specific locations). Implemented with locationID as FK in Rentals. Each rental is associated with exactly one location.
 - Crows foot notation: Locations || - Rentals |<
- Locations to Vehicles : 1:M (See Vehicles section)
 - Crows foot notation : Location || - Vehicles |<
- Locations to Employees : 1:M (one location has many employees) Implemented with locationID as FK in employees
 - Crows foot notation : Locations || - Employees |<
- **Employees:** Records details of staff members who manage rentals and interact with clients. This entity tracks employees and their primary assigned location for operational management.
 - employeeID : int, auto_increment, unique, not NULL, PK
 - firstName: varchar(50), not NULL

- lastName: varchar(50), not NULL
- email : varchar(100), not NULL, unique
- phone : varchar(20)
- hireDate : date, not NULL
- role : varchar(30)
- locationID : int, not NULL, FK references Locations(locationID)

Relationships:

- Employees to Locations : M:1 (each employee is assigned to one primary location)
 - Crows foot notation : Locations || - Employees |<
- Employees to EmployeeClients : 1:M (One employee can assist many clients) Implemented with employeeID as FK in EmployeeClients.
 - Crows foot notation : Employees || - EmployeeClients |<
- **Rentals:** Records each individual rental transaction. This entity implements the many-to-many relationship between Clients and Vehicles while recording rental transaction details. This prevents overlapping bookings and supports revenue tracking.
 - rentalID: int, auto_increment, unique, not NULL, PK
 - clientID: int, not NULL, FK references Clients(clientID)
 - vehicleID: int, not NULL, FK references Vehicles(vehicleID)
 - locationID: int, not NULL, FK references Locations(locationID)
 - pickupDate: date, not NULL
 - returnDate: date, not NULL
 - actualReturnDate: date (nullable)
 - totalCost: decimal(10,2), not NULL
 - status: varchar(20) (e.g., "Active", "Completed", "Cancelled", "Overdue")

Relationships:

- Rentals to Clients: M:1 (see Clients section).
 - Crow's Foot Notation: Clients || — Rentals |<

- Rentals to Vehicles: M:1 (see Vehicles section).
 - Crow's Foot Notation: Vehicles || — Rentals |<
- Rentals to Locations: M:1 (see Locations section).
 - Crow's Foot Notation: Locations || — Rentals |<
- **EmployeeClients:** Table that implements the many-to-many relationship between Clients and Employees. This entity records which employees interacted with which clients and allows an employee to assist multiple clients while a client can work with multiple employees over time.
 - employeeClientID: int, auto_increment, unique, not NULL, PK
 - employeeID: int, not NULL, FK references Employees(employeeID)
 - clientID: int, not NULL, FK references Clients(clientID)

Relationships:

- EmployeeClients to Employees: M:1 (see Employees section).
 - Crow's Foot Notation: Employees || — EmployeeClients |<
- EmployeeClients to Clients: M:1 (see Clients section).
 - Crow's Foot Notation: Clients || — EmployeeClients |<

Schema Diagram

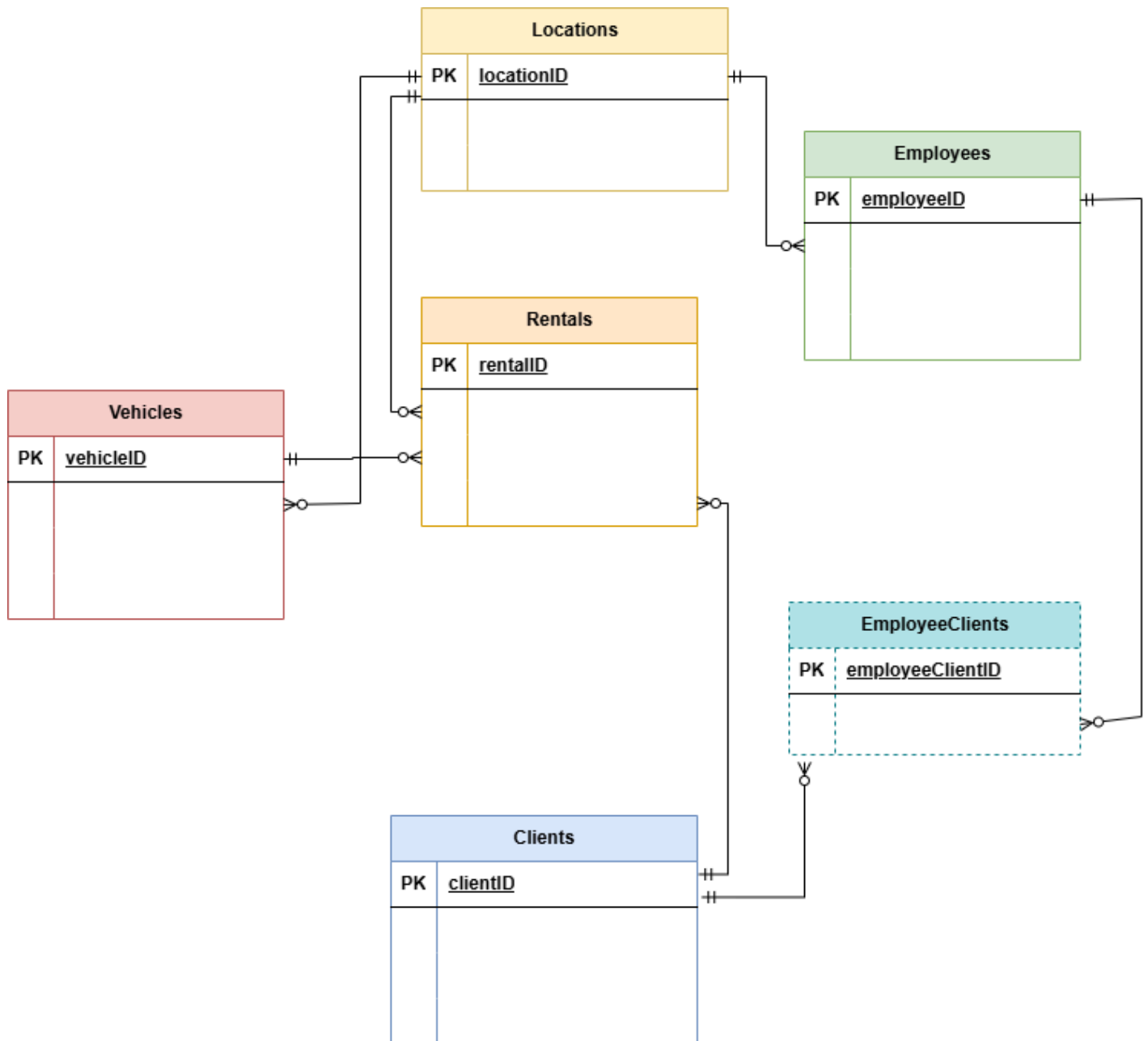
Normalization Process

We created a sample rental report that contained repeated client names, vehicle details, employee names, and location information in every row. This caused significant redundancy and update/insertion/deletion anomalies.

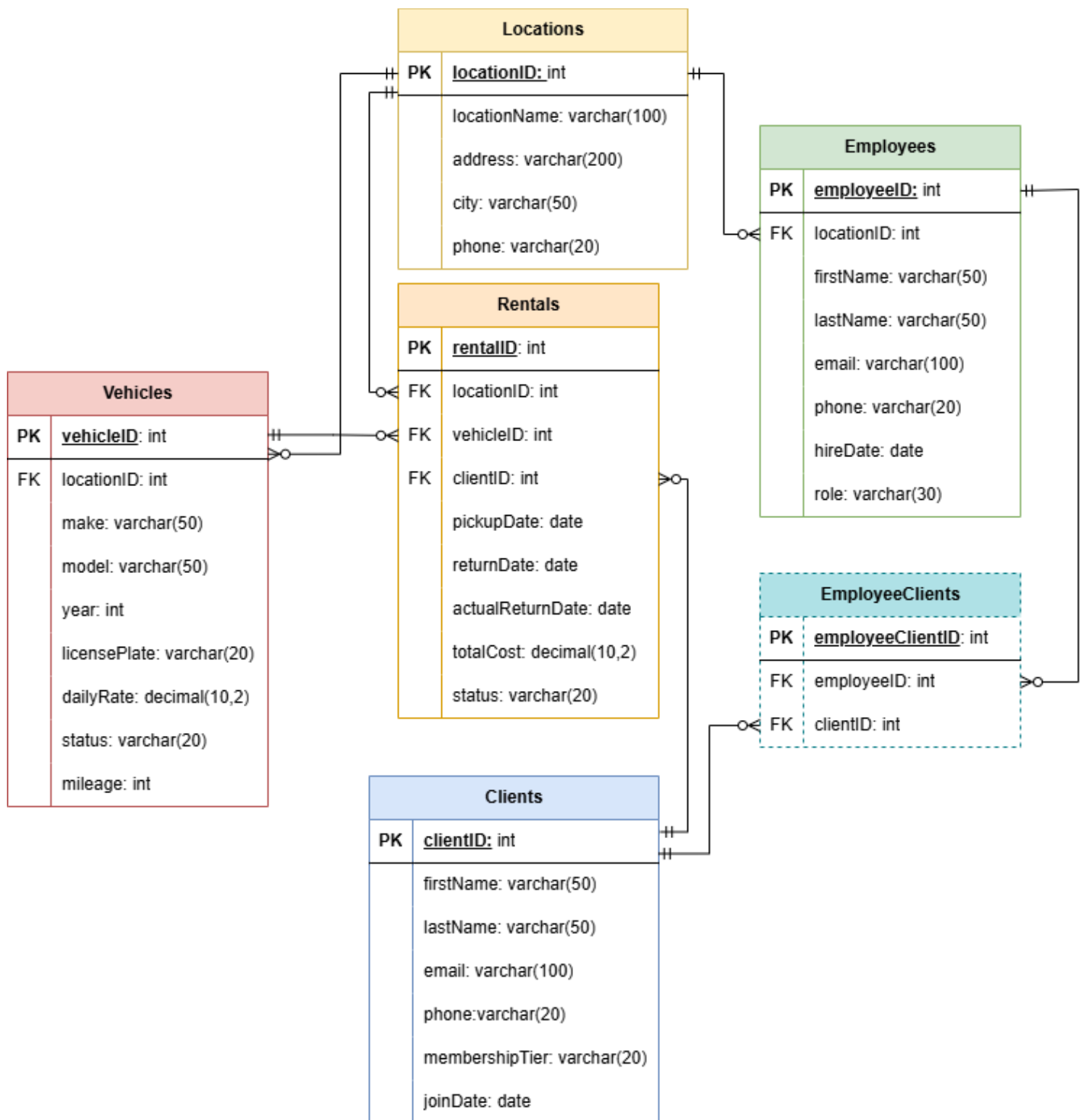
- 1NF: All attributes are atomic; no multi-valued or repeating groups exist.
- 2NF: Surrogate primary keys (auto_increment IDs) were used for every table, ensuring every non-key attribute depends on the entire primary key and eliminating partial dependencies.
- 3NF: No transitive dependencies remain. All client data lives only in the Clients table, vehicle status and details only in the Vehicles table, employee assignment only in the Employees table, etc. The totalCost field was intentionally kept in the Rentals table for historical accuracy and performance (a standard and acceptable practice for transaction tables).

The final schema is in 3NF and matches the updated Database Outline and ERD exactly.

ERD Diagram:



Schema:



Example Data

The data demonstrates all relationships in action (one client with multiple rentals, vehicles in different statuses, employees assisting multiple clients via the intersection table, rentals across multiple locations, etc.).

Locations

locationID	locationName	address	city	phone
1	Beverly Hills	123 Rodeo Drive	Beverly Hills	(310) 555-0101
2	Newport Beach	456 Pacific Coast Hwy	Newport Beach	(949) 555-0202
3	Irvine	789 Jamboree Road	Irvine	(949) 555-0303

Clients

clientID	firstName	lastName	email	phone	membershipTier	joinDate
1	Emma	Thompson	emma.t@email.com	(310) 555-1001	VIP	2024-01-15
2	Michael	Rodriguez	michael.r@email.com	(949) 555-1002	Premium	2024-03-20
3	Sophia	Chen	sophia.c@email.com	(714) 555-1003	Standard	2025-02-10
4	James	Patel	james.p@email.com	(310) 555-1004	VIP	2023-11-05

Vehicles

vehicleID	locationID	make	model	year	licensePlate	dailyRate	status	mileage
1	1	Ferrari	488 Spider	2023	FERR488	1499.99	Available	2450
2	1	Lambo rghini	Urus	2024	LAMBU24	1299.99	Rented	1800
3	2	Rolls Royce	Ghost	2023	RRGHOST	1799.99	Available	3200
4	3	Porsch e	911 Turbo S	2024	P911TS	899.99	Mainten ance	890
5	2	Bentley	Continent al GT	2023	BENT23	1599.99	Available	1450

Employees

employeeID	firstName	lastName	email	phone	hireDate	role	locationID
1	David	Kim	david.k@elitrentals.com	(310) 555-2001	2023-06-01	Manager	1
2	Olivia	Martinez	olivia.m@elitrentals.com	(949) 555-2002	2024-01-15	Rental Specialist	2
3	Liam	Nguyen	liam.n@elitrentals.com	(949) 555-2003	2024-09-01	Service Advisor	3
4	Isabella	Garcia	isabella.g@elitrentals.com	(310) 555-2004	2023-11-10	Rental Specialist	1

Rentals

renta IID	clien tID	vehicle ID	locatio nID	pickupDat e	returnDat e	actualRetur nDate	totalCost	status
1	1	2	1	2025-04-01	2025-04-05	NULL	5199.96	Active
2	2	1	1	2025-03-15	2025-03-18	2025-03-18	4499.97	Complete d
3	3	5	2	2025-04-10	2025-04-12	NULL	3199.98	Active
4	4	3	2	2025-02-20	2025-02-25	2025-02-25	8999.95	Complete d
5	1	4	3	2025-01-05	2025-01-07	2025-01-07	1799.98	Complete d

EmployeeClients (M:N intersection table)

employeeClientID	employeeID	clientID
1	1	1
2	1	2
3	2	3
4	2	4
5	4	1
6	4	3
7	3	2

Citations

No external sources were used.